

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 03

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Recap of Previous Lecture



Topic

Synchronous Pipeline

Topic

Latency & Throughput

Topic

Instruction Pipelining



Topics to be Covered



Topic

Instruction Pipelining

Topic

Pipeline Hazards

Topic

Operand Forwarding



Clock Cycles

[illegible]



Topic : Pipeline Hazards



Situations that prevent the next instruction from being executing during its designated clock cycle





Topic : Pipeline Hazards

1. Structural Hazard / Resource Conflict
2. Data Hazard / Data Dependency
3. Control Hazard / Branch Difficulty





Topic : Structural Hazard

2 different instructions try to use same resource at same time.





Topic : Data Hazard or Data Dependency

Result of an instruction is used as input in next

$R1 \leftarrow R2 + R3$

IF ID OF EX WB

$R5 \leftarrow R1 * R6$

IF ID — — OF EX WB



Solⁿ of data dependency

s/w solⁿ
(by compiler)
⇓
delayed load

$R1 \leftarrow R2 + R3$

} independent or
no-operatⁿ
inst^{ns}

$R5 \leftarrow R1 * R6$

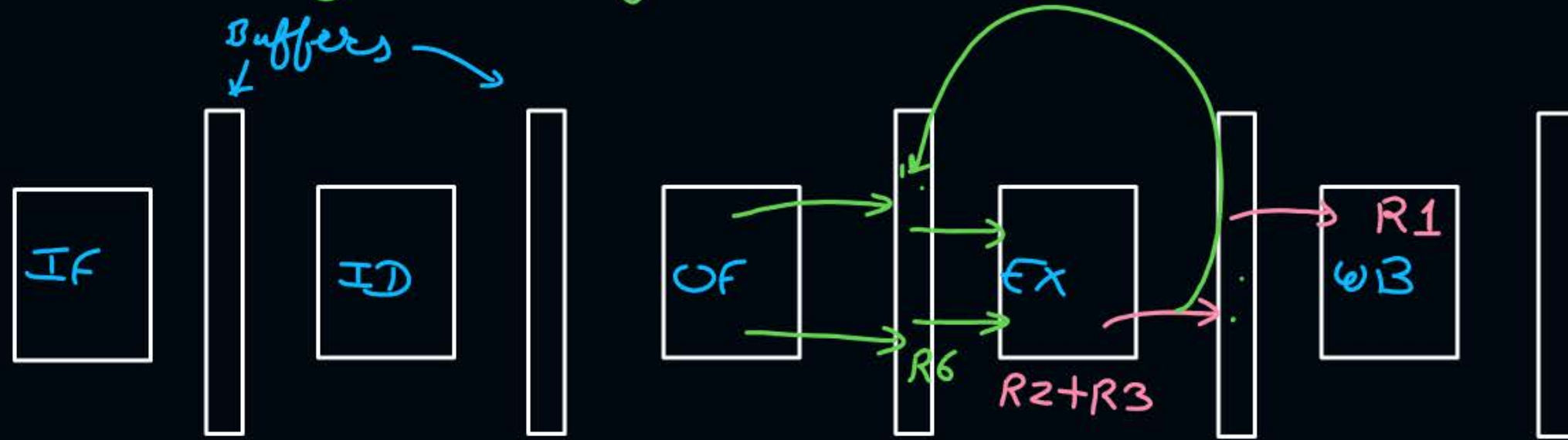
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB

h/w solⁿ

→ H/w interlocking
→ operand forwarding

putting stall
cycles.

operand forwarding / bypassing :-



$R1 \leftarrow R2+R3$

$R1 * R6$

IF ID OF EX WB
IF ID OF EX WB
 $\nearrow$
 $\searrow$
 R6

} no stalls for
data dependency
from ALU to
ALU.

Load to ALU:-

$R1 \leftarrow m[\text{add.}]$

$R5 \leftarrow R1 * R6$

ALU to store:-

$R1 \leftarrow R2 + R3$

$m[1000] \leftarrow R1$

} no saving in stall even if operand forw.
is used

}

no. of stalls due to data dependency = WB phase no. - OF phase no.

$$= 5 - 3$$

$$= 2$$

$R1 \leftarrow R2 + R3$

$R7 \leftarrow R8 + 1$

$R5 \leftarrow R1 * R6$

no. of stalls due to data dependency = $\left(\text{WB phase no.} - \text{OF phase no.} \right) - \text{no. of independent instn b/w them}$



Topic : Control Hazard or Branch Difficulty

Hazards because of branch instructions

solⁿ

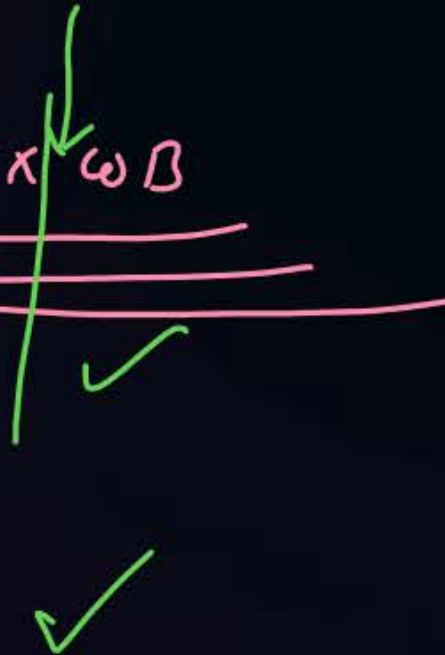
slow solⁿ
(by compiler)

Delayed branch

branch instⁿ

Next instⁿ

IF ID OF EX WB



H/w solⁿ

↓
Branch prediction

Ans = 8

#Q. Consider a pipelined processor with the following four stages:

- 1 IF: Instruction Fetch
- 2 ID: Instruction Decode and Operand Fetch
- 3 EX: Execute
- 4 WB: Write Back

The IF, ID, and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycle in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD R2, R1, R0

MUL R4, R3, R2

SUB R6, R5, R4

$R2 \leftarrow R1 + R0$

$R4 \leftarrow R3 * R2$

$R6 \leftarrow R5 - R4$

IF	1			
ID	1	ADD	SUB	MUL
EX		1	1	3
WB	1	Stalls	0	0
				2

#Q. In previous question if operand forwarding is not used then What is the number of clock cycles taken to complete the following sequence of instructions?

$$\text{no. of stalls per data dependency} = 4 - 2 = 2$$

$$\text{w/o hazard} = 4 + 3 - 1 = 6$$

$$\text{stalls for structural hazard} = 2$$

$$\text{---||--- data ---||---} = 2 * 2 = 4$$

$$\underline{\underline{12}} \text{ Ans.}$$

Speed up of using operand forwarding as compared to not using — ?

$$= \frac{12}{8}$$

$$= 1.5$$

#Q. In previous question if 10 ADD type, 12 MUL type and 7 SUB type instructions are there then What is the number of clock cycles taken to complete the following sequence of instructions?

$$n = 10 + 12 + 7 = 29$$

$$\text{w/o stalls cycles} = 4 + 29 - 1 = 32$$

$$\text{stalls for structural hazard} = (12 * 2) = 24$$

$$\text{Ans} = \underline{\underline{56}}$$

#Q. Consider the sequence of machine instruction given below:

MUL R5, R0, R1

DIV R6, R2, R3

ADD R7, R5, R6

SUB R8, R7, R4

w/o hazard = $4 + 4 - 1 = 7$

stalls

= $2 + 4 = 6$

13 Ans.

In the above sequence, R0 to R8 are general purpose registers. In the instructions shown, the first register shows the result of the operation performed on the second and the third registers. This sequence of instructions is to be executed in a pipelined instruction processor with the following stages: (1) Instruction Fetch and Decode (IF), (2) Operand Fetch (OF), (3) Perform Operation (PO), and (4) Write back the result (WB). The IF, OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses operand forwarding from the PO stage to the OF stage. The number of clock cycles taken for the execution of the above sequence of instruction is _____.

IF	1				
OF	1				
PO		ADD	SUB	MUL	DIV
WB	1	1	1	3	5
		Stalls	0	0	2
					4

#Q. In previous question if 7 ADD type, 4 MUL type, 5 DIV type and 6 SUB type instructions are there then What is the number of clock cycles taken to complete the following sequence of instructions?

$$n = 22$$

$$\text{w/o stalls} = 4 + 22 - 1 = 25$$

$$\text{stalls} = \frac{(4 * 2) + (5 * 4) = 28}{\text{Total} = \underline{\underline{53}} \text{ Ans.}}$$

#Q. The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instruction is _____?

Ans = 23

#Q. Consider a 4-stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below:

	S ₁	S ₂	S ₃	S ₄
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

What is the number of cycles needed to execute the following loop?

For (i = 1 to 2) (I1; I2; I3; I4)

[illegible]

[NAT]



	S_1	S_2	S_3	S_4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

	S1	S2	S3	S4
I1	2	3	4	5
I2	3	6	8	10
I3	5	7	9	13
I4	6	9	11	15
I1	8	10	12	16
I2	9	13	15	18
I3	11	14	16	21
I4	12	16	18	23



Data



Topic : Hazard Classification

Assume that there are two instructions i and j , and i is executed before j

RAW

WAW

WAR





Topic : Read After Write (RAW)

→ True dependency

j tries to read a source before i writes it. So j incorrectly gets the old value.

$$\left. \begin{array}{l} R1 \leftarrow R2 + R3 \\ R5 \leftarrow R1 * R6 \end{array} \right\} \underline{\text{Sol}^n} :- \text{operand forwarding}$$



Topic : Write After Write (WAW)

→ write dependency
→ false dependency } no stalls ever

j tries to write an operand before it is written by i

$i: R1 \leftarrow R2 + R3$
 $j: R1 \leftarrow R5 * R6$ } Solⁿ:- Register Renaming (H/w Solⁿ)

$R1 \leftarrow R2 + R3$
 $R9 \leftarrow R5 * R6$



Topic : Write After Read (WAR)

→ anti-dependency } no stalls ever
→ false "

j tries to write an operand before i reads it, hence i incorrectly gets new value

$i:$ $R1 \leftarrow R2 + R3$
 $j:$ $R2 \leftarrow R6 * R5$ } solⁿ Reg. Renaming
↓
 $R1 \leftarrow R2 + R3$
 $R9 \leftarrow R6 * R5$

#Q. Count the number of RAW, WAW and WAR dependencies?

ADD R2, R1, R0

MUL R4, R3, R2

SUB R6, R5, R4

ADD R6, R7, R8

MUL R7, R1, R2

SUB R1, R3, R4

R2 ← R1 + R0

R4 ← R3 * R2

R6 ← R5 - R4

R6 ← R7 + R8

R7 ← R1 * R2

R1 ← R3 - R4

4 1 3

Ques)

$$R1 \leftarrow R2 + R3$$

$$R5 \leftarrow R1 * R4$$

$$R1 \leftarrow R9 - R6$$

$$R2 \leftarrow R1 + R8$$

no. of RAW dependencies

2 }

#Q. An instruction format has the following structure

Instruction number: opcode destination reg, source reg-1, source reg-2

Consider the following sequence of instruction to be executed in a pipelined processor

I1: DIV R3, R1, R2

I2: SUB R5, R3, R4

I3: ADD R3, R5, R6

I4: MUL R7, R3, R8

Which of the following statement is/are TRUE?

- A** There is a WAW dependency on R3 between I3 and I4
- B** There is a WAW dependency on R3 between I1 and I3
- C** There is a RAW dependency on R3 between I1 and I2
- D** There is a RAW dependency on R3 between I2 and I3

#Q. Consider the following sequence of instructions, which is to be executed on a 5 stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding
2. When pipeline does not use operand forwarding

ADD R2, R1, R0

$R2 \leftarrow R1 + R0$

MUL R4, R3, R2

$R4 \leftarrow R3 * R2$

SUB R6, R5, R4

$R6 \leftarrow R5 - R4$

ADD R6, R7, R8

$R6 \leftarrow R7 + R8$

MUL R7, R1, R2

$R7 \leftarrow R1 * R2$

SUB R1, R3, R4

$R1 \leftarrow R3 - R4$

#Q. Consider the following sequence of instructions, which is to be executed on a 5 stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding
2. When pipeline does not use operand forwarding

ADD R2, R1, R0	$R2 \leftarrow R1 + R0$
MUL R4, R3, R2	$R4 \leftarrow R3 * R2$
LOAD R7, (1000)	$R7 \leftarrow M[1000]$
ADD R6, R7, R8	$R6 \leftarrow R7 + R8$
MUL R7, R1, R2	$R7 \leftarrow R1 * R2$
SUB R1, R3, R4	$R1 \leftarrow R3 - R4$



2 mins Summary



Topic

Instruction Pipelining

Topic

Pipeline Hazards

Topic

Operand Forwarding





Happy Learning

THANK - YOU

